

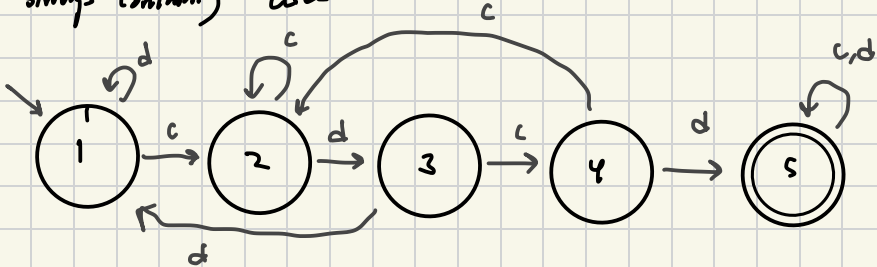
1. Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts, the alphabet is $\{c, d\}$. (15 points)

- (a) $\{w \mid w \text{ contains } cdcd \text{ as a substring}\}$ with 5 states.
- (b) The language $\{c\}$ with two states.
- (c) The language $\{\epsilon\}$ with one state.
- (d) $\{w \mid w \text{ does not contain a } d\}$ with one state.
- (e) $\{w \mid w \text{ contains an even number of } c\text{'s, or exactly two } d\text{'s}\}$ with 6 states.

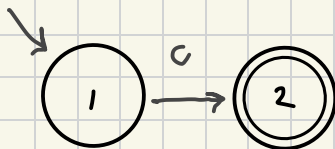
Matt Krueger
HW #2

For the last part above, it may be helpful to wait and see how we can use NFAs to show that the union of two regular languages is regular.

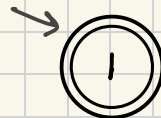
a) strings containing $cdcd$



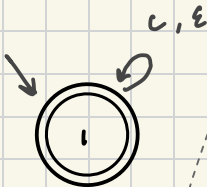
b) string c



c)

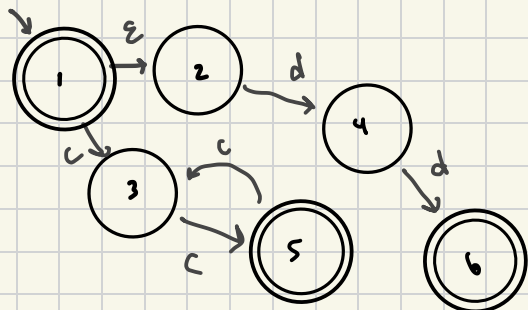


d) c^*



e) even # of c 's or 2 d 's

$\{\epsilon, dd, cc, cccc, ccccd, \dots\}$



2. Show by giving an example that if M is an NFA that recognizes language C , swapping the accept and non-accept states of M doesn't necessarily yield an NFA that recognizes the complement of C . (5 points)

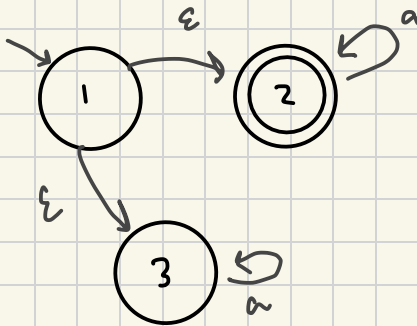
M : recognizes C

$$C = \{a, b\}^*$$

M' : may or may not recognize

Example:

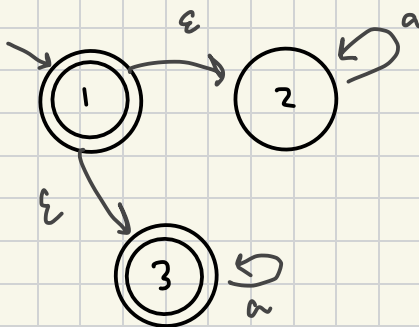
M



M accepts a^*

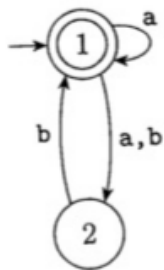
b^*

M'



M' accepts a^*

3. Convert the NFA in Figure 1 into a DFA using the systematic procedure that we studied. (7 points)



procedure:

1. $Q' = P(Q)$:

Every state of M is a set of states of N

$P(Q)$ is the set of subsets of Q

2. for $R \in Q'$ and $a \in \Sigma$, let $\delta'(R, a) = \{q \in Q \mid q \in \delta(r, a) \text{ for some } r \in R\}$

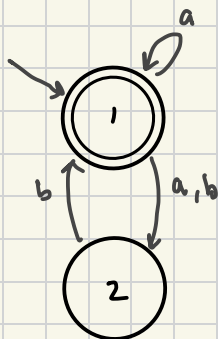
i.e. $\delta'(R, a) = \bigcup_{r \in R} \delta(r, a)$

3. $q'_0 = \{q_0\} \rightarrow \{1\}$

4. $F' = \{R \in Q' \mid R \text{ contains an accept state of } N\}$

1

set of states = $\{\emptyset, \{1\}, \{2\}, \{1,2\}\}$



$N = (Q, \Sigma, \delta, q_0, F)$

$Q = \{1, 2\}$

$\Sigma = \{a, b\}$

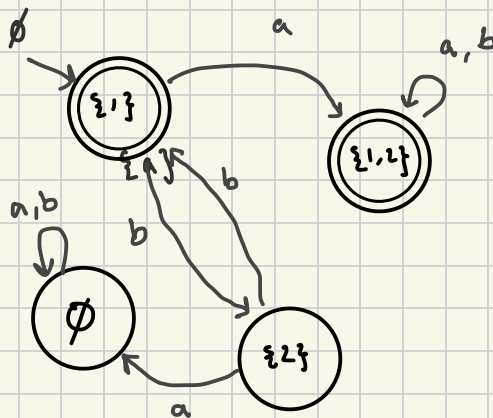
$\delta =$

	a	b	ϵ
1	$\{1, 2\}$	$\{2\}$	$\emptyset$
2	$\emptyset$	$\{1\}$	$\emptyset$

start

$q_0 = 1$

$F = \{1\}$



$M = (Q', \Sigma, \delta, q'_0, F')$

$Q' = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$

$\Sigma = \{a, b\}$

$\delta' =$

	a	b
$\emptyset$	$\emptyset$	$\emptyset$
$\{1\}$	$\{1, 2\}$	$\{2\}$
$\{2\}$	$\emptyset$	$\{1\}$
$\{1, 2\}$	$\{1, 2\}$	$\{1, 2\}$

$q'_0 = \{1\}$

$F' = \{\{1\}, \{1, 2\}\}$

4. For any string $w = w_1 w_2 \dots w_n$, the reverse of w , written w^R , is the string w in reverse order; that is, $w^R = w_n \dots w_2 w_1$. For any language A , let $A^R = \{w^R \mid w \in A\}$. Show that if A is regular, so is A^R . (8 points)

$$\begin{array}{ccc} A & & A^R \\ w = w_1 w_2 \dots w_n & \xrightarrow{\text{reverse}} & w_n \dots w_2 w_1 = w^R \end{array}$$

$$A^R = \{w^R \mid w \in A\}$$

claim: if A is Regular, then A^R is regular

PROOF: closure under reversal

Given that M accepts language A with alphabet Σ (DFA)
we can build a new machine

N that accepts A^R , the reversal of A with same Σ (NFA)

$$M = (Q, \Sigma, \delta, q_0, F)$$

$\rightarrow$

$$N = (Q', \Sigma, \delta', q_0', F')$$

$$Q \rightarrow Q \cup \{q_\epsilon\} \quad \text{where } q_\epsilon \text{ is new}$$

$$\Sigma \rightarrow \Sigma$$

$$q_0 \rightarrow q_\epsilon$$

$$F \rightarrow \{q_0\}$$

$$\delta \rightarrow \delta'$$

as this is reversed string

$\forall p, q \in Q$ and

$\forall a \in \Sigma$, if

$\delta(q, a) = p$ then N

contains transition from

$p \rightarrow q$ with trigger 'a'

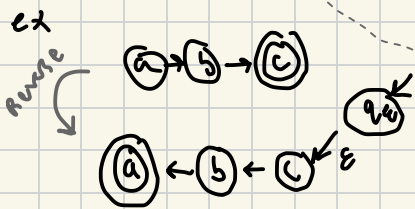
add $q_\epsilon \rightarrow F$
for all $F \in F$
with trigger ϵ .

(ex $\text{rev}(b) = b$)

$\therefore$ Thus $N(M^R)$ has part

from M 's F to M 's q_0 with

any former accept state accounted for. Because NFAs accept regular languages N accepts A^R , which is regular.



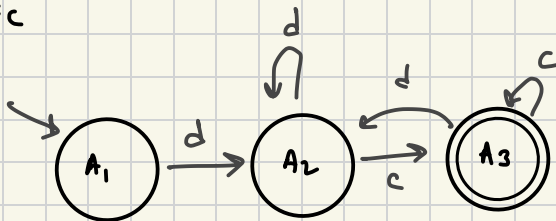
5. In class, we saw simple constructions using NFAs to show that regular languages are closed under the union, concatenation, and star operations. Use these constructions to answer this and the next two questions. The alphabet in these questions is $\{c, d\}$.

Give the state diagram of an NFA recognizing the union of the languages

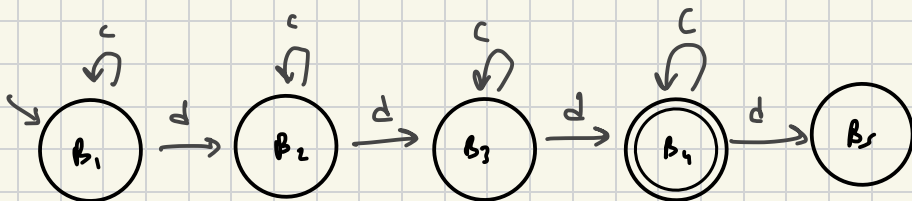
$A = \{w \mid w \text{ begins with a } d \text{ and ends with a } c\}$, and

$B = \{w \mid w \text{ contains at least three } d\text{'s}\}$. (5 points)

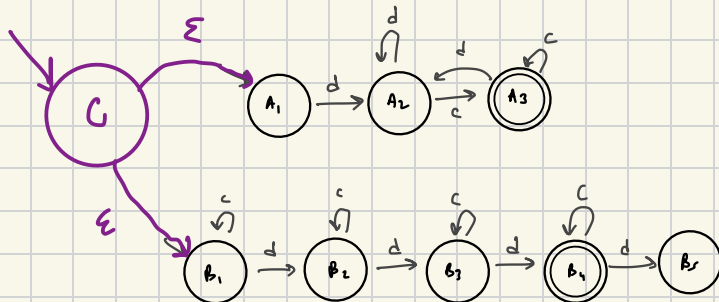
A: $d \Sigma^* c$



B: $(\Sigma^* d \Sigma^*) (\Sigma^* d \Sigma^*) (\Sigma^* d \Sigma^*)$



$A \cup B$: NFA



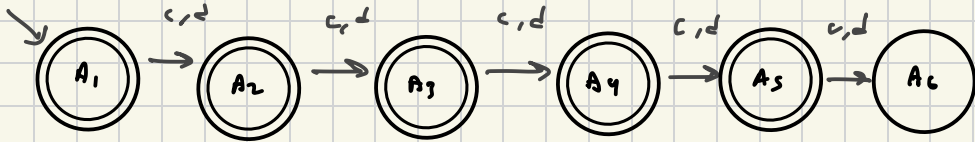
6. Give the state diagram of an NFA recognizing the concatenation of the languages

$$A = \{w \mid \text{the length of } w \text{ is at most } 5\}, \text{ and}$$

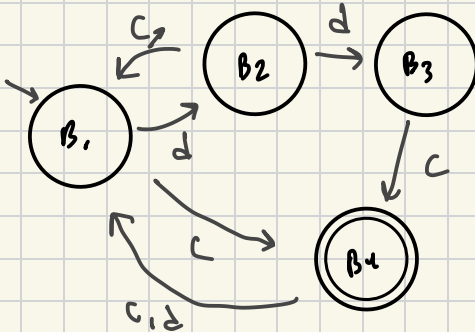
$$B = \{w \mid \text{every odd position in } w \text{ is a } c\}. \quad (5 \text{ points})$$

A:

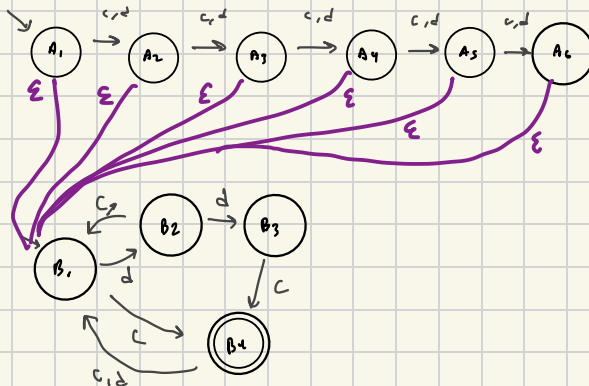
$$\Sigma = \{c, d\}$$



B:



A ∘ B:

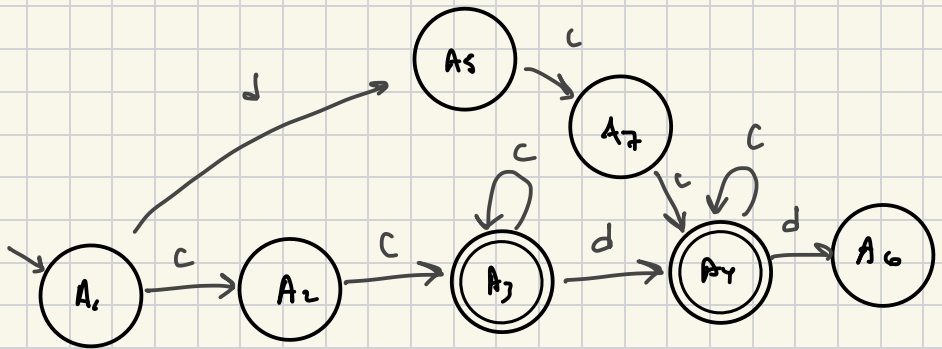


7. Give the state diagram of an NFA recognizing the star of the language

$A = \{w \mid w \text{ contains at least two } c\text{'s and at most one } d\}$. (5 points)

$$\Sigma = \{c, d\}$$

$$A = (cc^+d) \cup (dcc^+) \cup (c^+dc^+) \cup (cc^+)$$



$$A^* = (cc^+d)^* \cup (dcc^+)^* \cup (c^+dc^+)^* \cup (cc^+)^*$$

